

Distribution of Random Variables

The Normal Distribution

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(openintro)
```

```
Loading required package: airports
Loading required package: cherryblossom
Loading required package: usdata
```

```
data("fastfood", package='openintro')
head(fastfood)
```

```
# A tibble: 6 x 17
  restaurant item      calories cal_fat total_fat sat_fat trans_fat cholesterol
  <chr>        <chr>      <dbl>  <dbl>    <dbl>  <dbl>    <dbl>      <dbl>
1 Mcdonalds   Artisan G~    380    60      7      2      0      95
```

2	Mcdonalds	Single Ba~	840	410	45	17	1.5	130
3	Mcdonalds	Double Ba~	1130	600	67	27	3	220
4	Mcdonalds	Grilled B~	750	280	31	10	0.5	155
5	Mcdonalds	Crispy Ba~	920	410	45	12	0.5	120
6	Mcdonalds	Big Mac	540	250	28	10	1	80

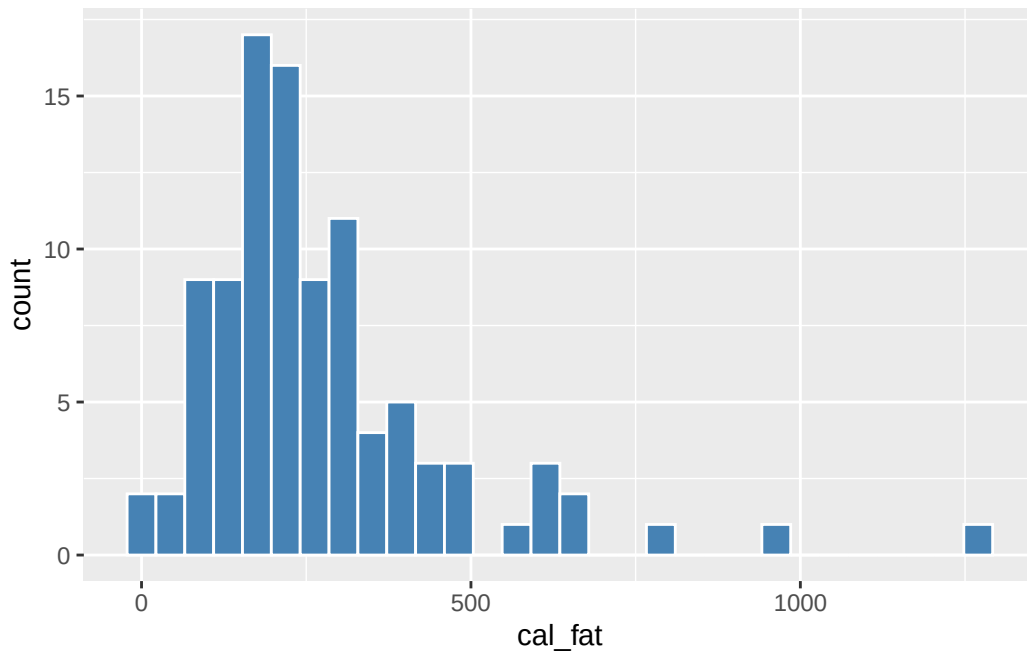
i 9 more variables: sodium <dbl>, total_carb <dbl>, fiber <dbl>, sugar <dbl>,
protein <dbl>, vit_a <dbl>, vit_c <dbl>, calcium <dbl>, salad <chr>

```
mcdonalds <- fastfood |>
  filter(restaurant == "Mcdonalds")
dairy_queen <- fastfood |>
  filter(restaurant == "Dairy Queen")
```

Exercise 1

```
fat_calories <- fastfood |>
  filter(restaurant %in% c("Mcdonalds", "Dairy Queen"))

ggplot(fat_calories, aes(x = cal_fat)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white")
```

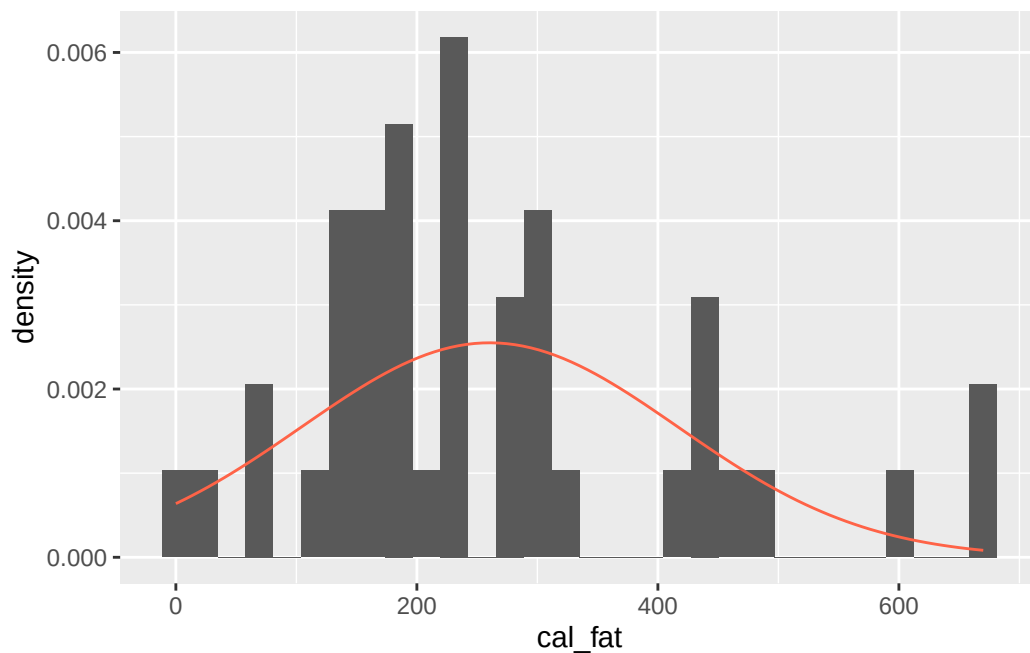


```
dqmean <- mean(dairy_queen$cal_fat)
dqsd   <- sd(dairy_queen$cal_fat)

ggplot(data = dairy_queen, aes(x = cal_fat)) +
  geom_blank() +
  geom_histogram(aes(y = ..density..)) +
  stat_function(fun = dnorm, args = c(mean = dqmean, sd = dqsd), col = "tomato")
```

Warning: The dot-dot notation (`..density..`) was deprecated in ggplot2 3.4.0.
 i Please use `after_stat(density)` instead.

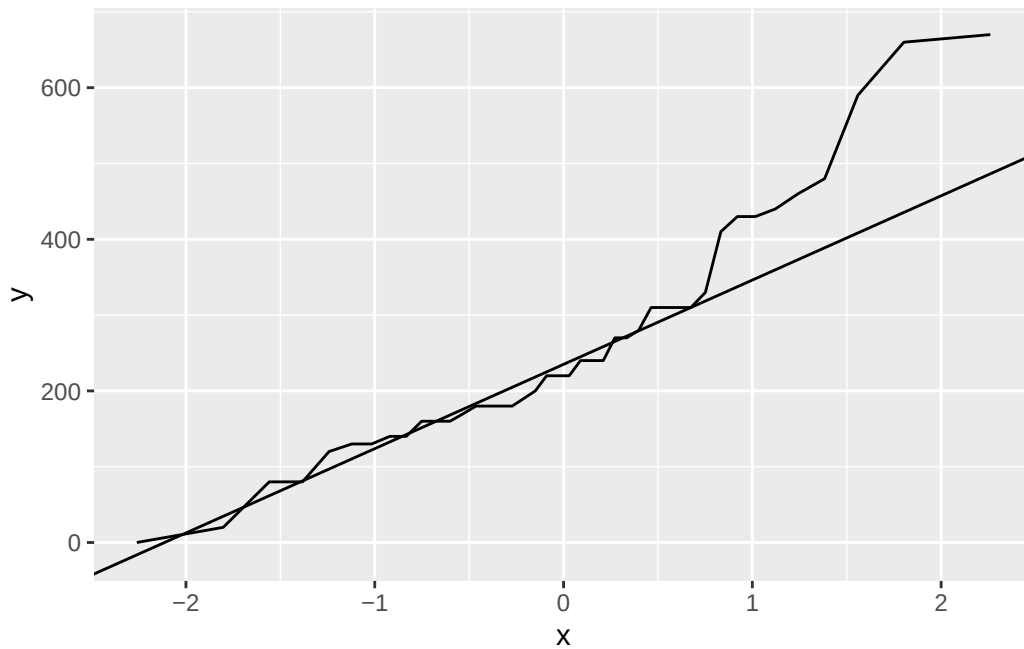
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.



Exercise 2

Based on the line the data does seem to follow a close to normal distribution.

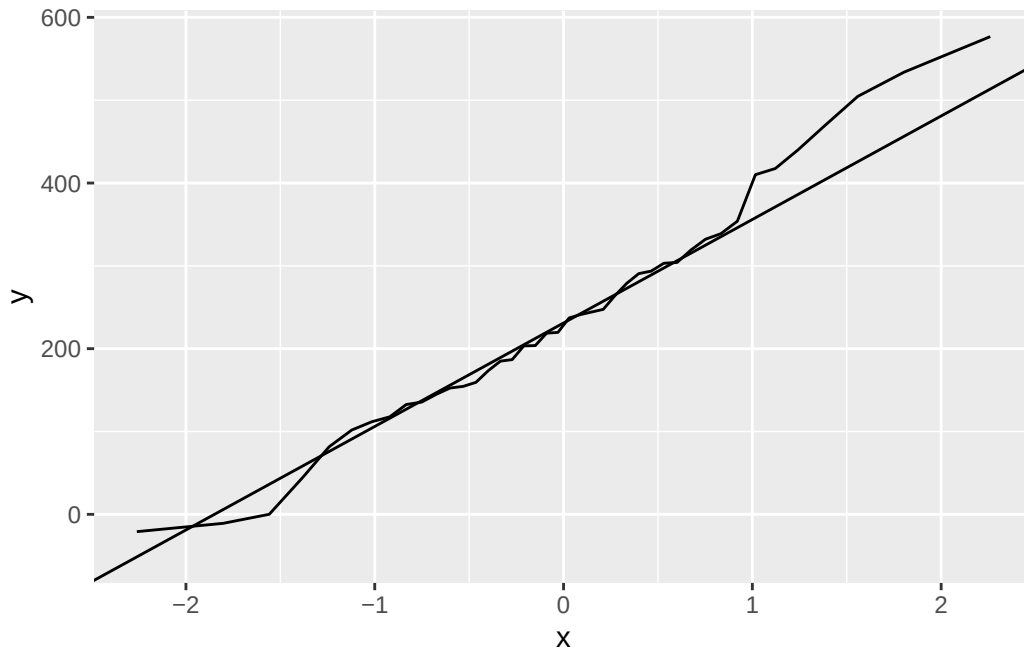
```
ggplot(data = dairy_queen, aes(sample = cal_fat)) +
  geom_line(stat = "qq") +
  stat_qq_line()
```



```
sim_norm <- rnorm(n = nrow(dairy_queen), mean = dqmean, sd = dqsd)
```

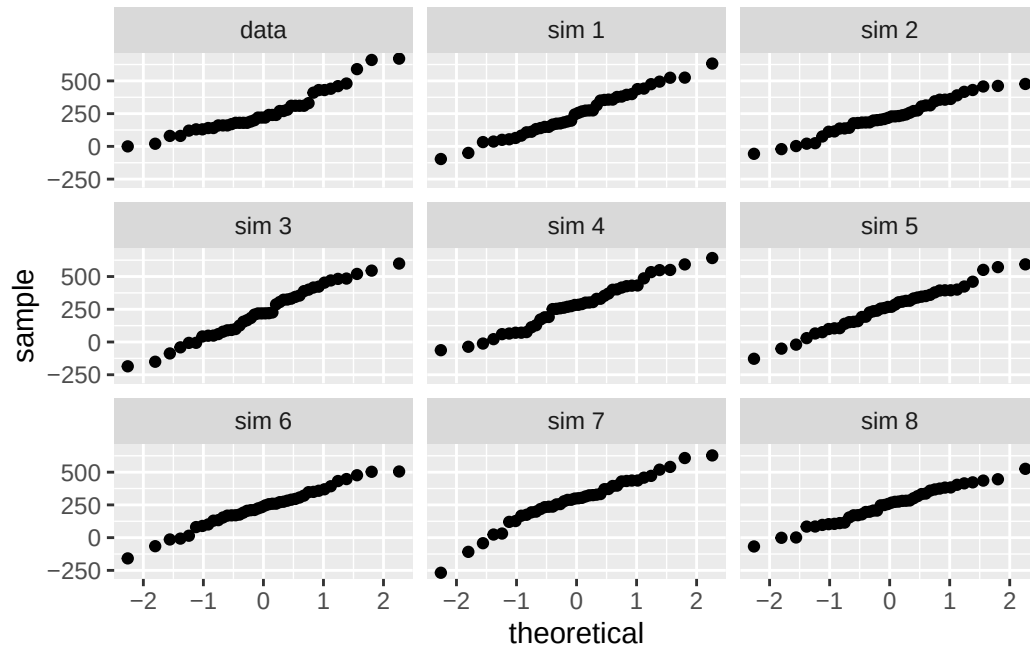
Exercise 3

```
ggplot( data = NULL, aes(sample = sim_norm))+  
  geom_line(stat = "qq") +  
  stat_qq_line()
```



```
qqnormsim(sample = cal_fat, data = dairy_queen)
```

Warning: `aes\_string()` was deprecated in ggplot2 3.0.0.
i Please use tidy evaluation idioms with `aes()`.
i See also `vignette("ggplot2-in-packages")` for more information.
i The deprecated feature was likely used in the openintro package.
Please report the issue at
<<https://github.com/OpenIntroStat/openintro/issues>>.

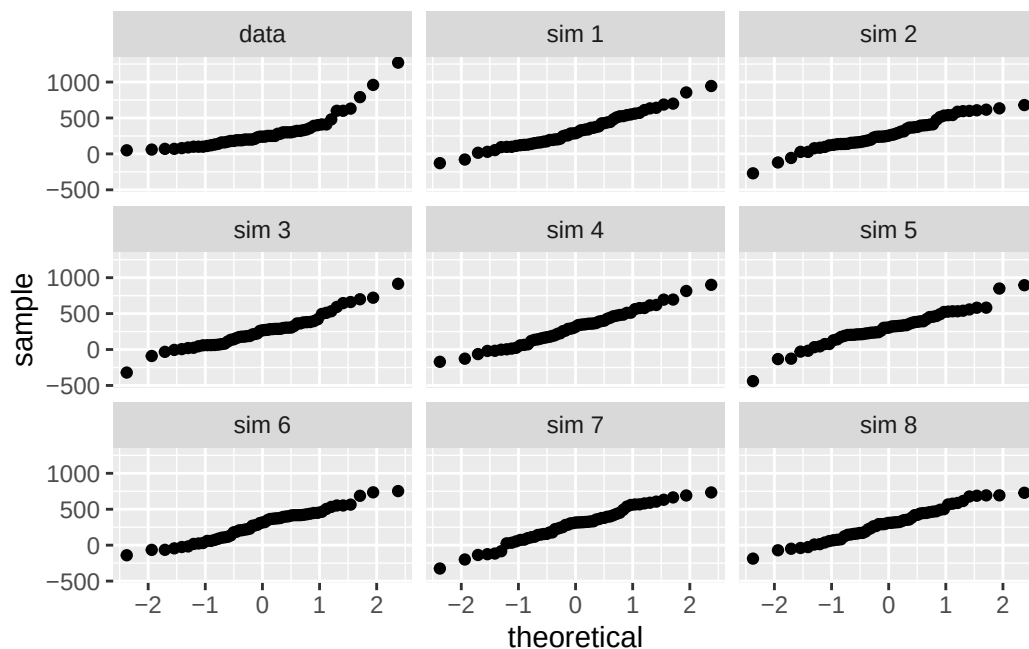


Exercise 4

Yes, the simulation looks similar to the plot created for the simulated data and brings us closer to showing that the the calories follow a normal distribution.

Exercise 5

```
qqnormsim( sample = cal_fat, data = mcdonalds)
```



Normal Probabilities

```
1 - pnorm(q = 600, mean = dqmean, sd = dqsd)
```

```
[1] 0.01501523
```

```
dairy_queen %>%
  filter(cal_fat > 600) %>%
  summarise(percent = n() / nrow(dairy_queen))
```

```
# A tibble: 1 x 1
  percent
  <dbl>
1 0.0476
```

Exercise 6

Two questions we can ask are what is the probability that a Taco Bell item is over 400 calories and what is the probability that an item at burger king is under 300 calories

```

taco <- fastfood %>% filter(restaurant == "Taco Bell")
bk    <- fastfood %>% filter(restaurant == "Burger King")

tb_mean <- mean(taco$calories)
tb_sd   <- sd(taco$calories)

q1_theoretical <- 1 - pnorm(400, mean = tb_mean, sd = tb_sd)
q1_empirical  <- mean(taco$calories > 400)

bk_mean <- mean(bk$cal_fat)
bk_sd   <- sd(bk$cal_fat)

q2_theoretical <- pnorm(300, mean = bk_mean, sd = bk_sd)
q2_empirical  <- mean(bk$cal_fat < 300)

```

The theoretical probability for taco bell is 0.5935926 while the emperical is 0.5565217 compared to the burger king theoretical and emperical number being 0.431106 ,0.5142857 respectively. The burger king probability is much closer to the theoretical probability

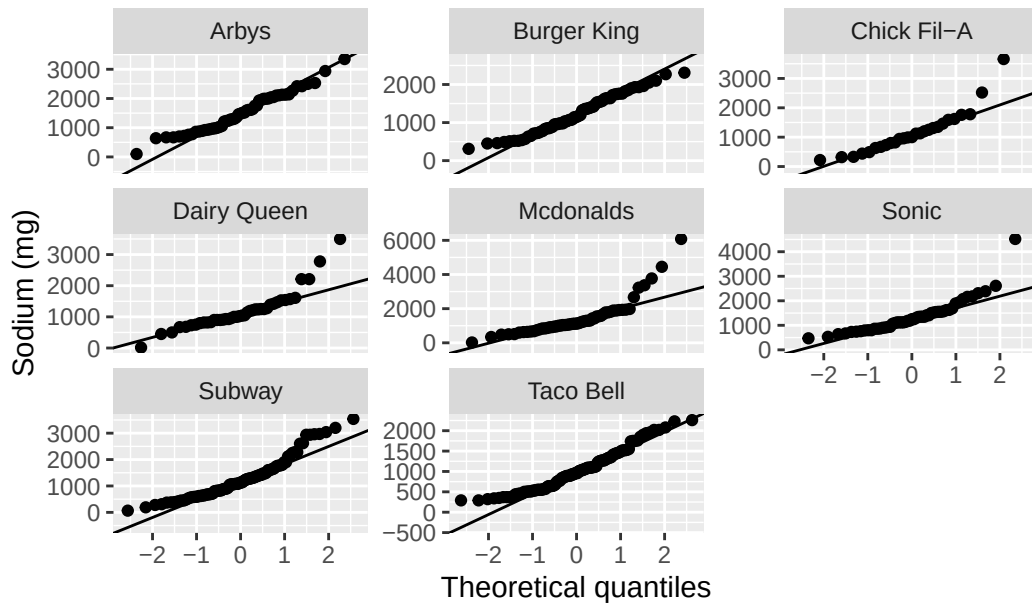
Exercise 7

```

ggplot(fastfood, aes(sample = sodium)) +
  geom_point(stat = "qq") +
  stat_qq_line() +
  facet_wrap(~ restaurant, scales = "free_y") +
  labs(title = "Normal probability plots of sodium by restaurant",
       x = "Theoretical quantiles", y = "Sodium (mg)")

```

Normal probability plots of sodium by restaurant



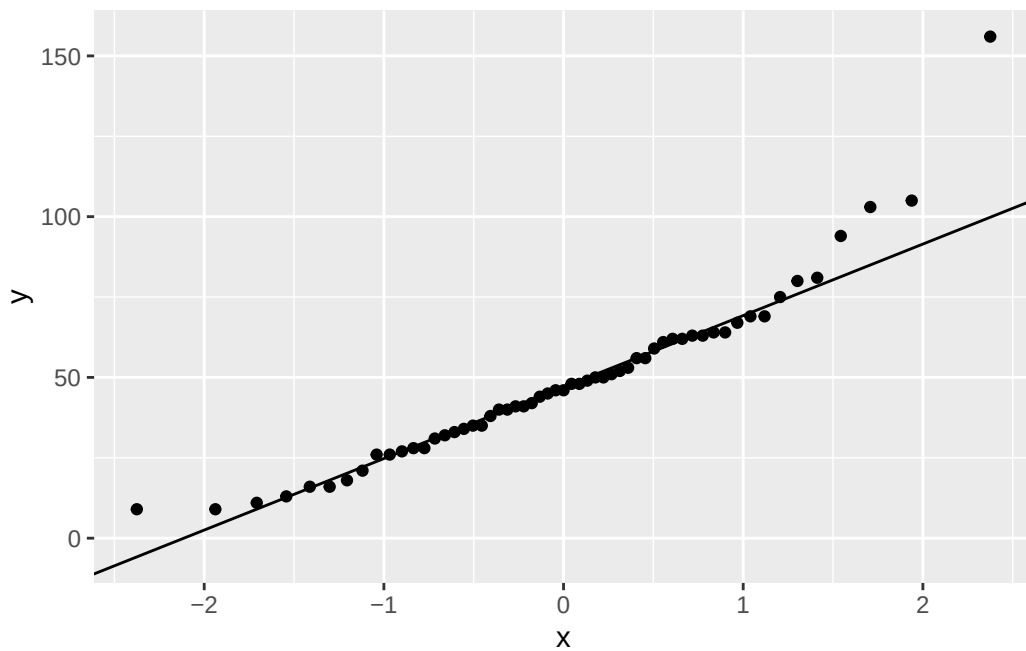
it is tough to tell which restaurant is the closest to normally distributed for sodium content, but it looks like Chick Fil-A is the nearest to the line with just two larger outliers. Ultimately, it may be Burger King that is the closest to normally distributed.

Exercise 8

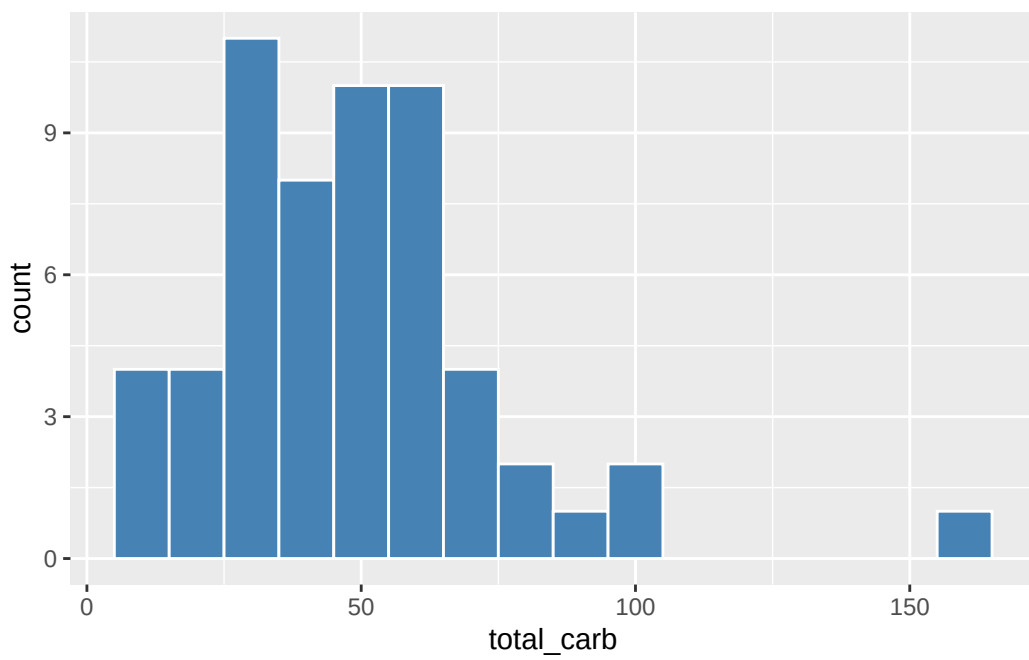
These distributions appear to be stepwise due to a uniform amount of salt being used in the recipe. Each restaurant is likely to use a number like 250 mgs as a unit for each increase in salt.

Exercise 9

```
ggplot(data = mcdonalds, aes(sample = total_carb)) +  
  geom_point(stat = "qq") +  
  stat_qq_line()
```



```
ggplot(data = mcdonalds, aes(x = total_carb)) +  
  geom_histogram(binwidth = 10, fill = "steelblue", color = "white")
```



It looks like the carbohydrate of mcdonalds is right skewed with most of the clustering being on the left side of the histogram.